# The Logical Starting Point of Mathematical Logic and Natural Generation

## Philosophical Ideas, Mathematical and Physical Pictures, and a Research Programme

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Abstract: We present the philosophical basis and research programme of a generative framework for mathematics and physics. The central question concerns the origin of the objects, assumptions, and operations from which a theory begins. Logical consistency constrains their consequences but does not establish their correspondence with nature. Taking minimal unbiased change as the common starting point, the framework studies how relations close into ordered structures and how successive levels acquire their own objects and laws. The main argument develops this view from the discussion in *Beyond Apparent Paths*. Three appendices develop the physical, mathematical, and engineering pictures that guide the programme. A common source connects internal spectra, gauge and matter structure, a positive equilibrium Yang–Mills gap, collective gravity, and cosmic evolution. Retaining the formation of numbers and functions supplies history-dependent responses, a signed scale-energy approach to the Riemann hypothesis, and a study of efficient generation and verification. For the Navier–Stokes equations, finite evolutions with controlled residuals, stability, and spectral tails are completed into continuum solutions on certified intervals. The unifying aim is to trace the starting points of these problems to a common process of generation and to retain that process in their mathematical description.

Keywords: Natural generation; Unified theories; Yang–Mills theory and mass gap; Navier–Stokes equations; $P$ versus $NP$; Riemann hypothesis; Mathematical foundations; Spectral geometry

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## 1 Introduction

A physical theory usually begins with fields, symmetries, and dynamical laws. A mathematical theory begins with objects, axioms, and permitted operations. These choices make rigorous reasoning possible, but leave open a prior question: where do the starting points themselves come from? The present article sets out the philosophical ideas and research programme of a generative framework built around this question. It explains why problems in particle physics, gravity, arithmetic, computational complexity, and nonlinear evolution are studied together, and how their treatment depends on a common account of generation.

The main argument develops the discussion of "The Starting Point of Logic" in *Jian Lu Bu Zou* and its English version, *Beyond Apparent Paths: The Practice of Seeking Truth from Facts* [1, 2]. Two kinds of starting point are distinguished. A local starting point specifies the objects, assumptions, operations, and conditions of a particular description. A common generative starting point concerns the formation of those objects and conditions. The distinction matters whenever a model is applied outside the level or range for which it was constructed.

The framework takes minimal unbiased change as the common origin. Disorder at this level means change before a structure has selected relations or directions. Relations, closure, and successive constraints then give rise to multi-level order. Formal reasoning describes what follows within a given level; emergent and dialectical reasoning concerns the formation of new levels and their relations. Observation and experiment continually test the resulting descriptions against nature. The mathematical task is to express generation through explicit operations and to retain the information needed when those operations are composed or passed to a limit.

The three appendices develop this programme beyond the general argument. Appendix A follows a common source into internal spectra, fields, collective gravity, and cosmic dynamics. Appendix B considers generated arithmetic and functions, the Riemann hypothesis, and the relation between generation and verification. Appendix C applies the same method to viscous evolution and its finite approximations. Figure 1 shows how these directions share a starting point while addressing different questions.

## 2 Logic, art and reality

Russell described pure mathematics as conditional reasoning about objects whose particular nature is left unspecified [3, p. 84]. Whether its premises describe something that exists is a further question. This distinction between the consequences of a starting point and its relation to reality also appears in familiar forms of storytelling.

Films, television dramas, and short videos generated by artificial intelligence routinely present events that we accept without difficulty: martial-arts masters leap across rooftops, spacecraft travel through time, and animated characters possess extraordinary powers. Each work establishes a world in which its story takes place. Once that setting is accepted, the characters and events can follow a coherent sequence of causes and consequences. The story has a logic of its own.

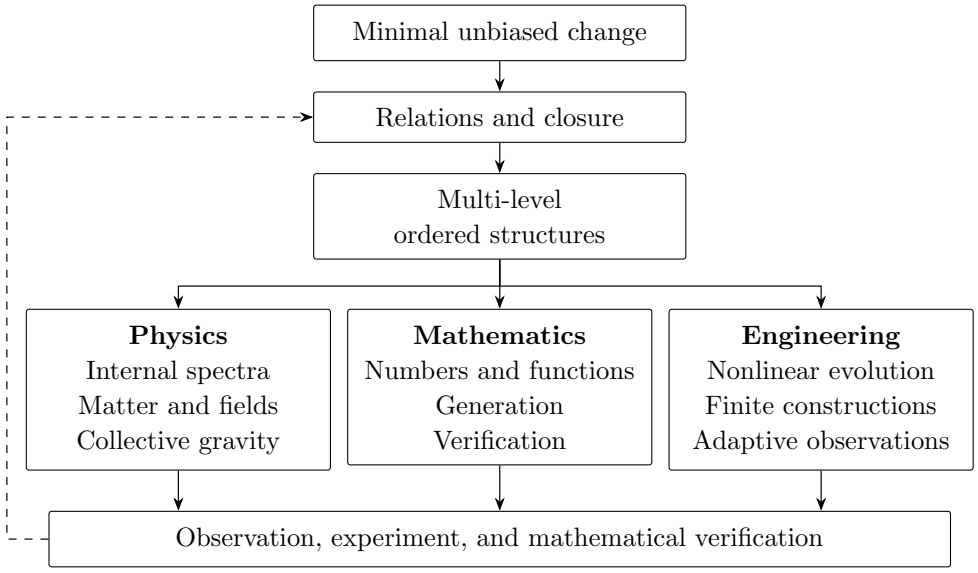



**Figure 1**. Organisation of the research programme. A common generative starting point leads to three related lines of inquiry. The dashed return path denotes correction of the descriptions and their local starting points through observation, experiment, and mathematical verification.

We nevertheless distinguish a coherent story from reality. Art can choose its premises freely; nature has its own conditions of generation and laws of change. Logic allows us to follow the consequences of a given starting point, but cannot establish that the starting point corresponds to nature. An error there can persist through an otherwise rigorous argument. The later steps may remain consistent while carrying the account progressively farther from what actually exists.

## 3 Nature's logic and multi-level order

Natural things must also conform to logic. All things are multi-level ordered structures, continually changing under the influence of disorder at the lowest level. A seed does not become a bird without cause, and atoms do not combine into matter arbitrarily. Disorder at the lowest level corresponds to absolute randomness. Within an ordered structure, this randomness is constrained at successive levels and appears through the probability laws of each level. The causal relations between changes constitute nature's own "logic". What we recognise as reasons and laws is the expression of these structural relations in cognition.

In the Preface to *Elements of the Philosophy of Right*, Hegel gave a well-known formulation of the relation between reason and reality. In H. B. Nisbet's translation, "What is rational is actual; and what is actual is rational" [4, Preface, p. 20]. Rationality here need not imply approval of everything that occurs. We understand it through the relations by which a thing arises, persists, and changes. These relations constitute its multi-level ordered structure; because they can be known, the thing appears intelligible and logical. On this view, logic belongs to the order of nature and can consequently be reflected in thought.

Logical consistency is therefore necessary for an account of reality, but it is not sufficient. A real thing has internal relations, whereas a logically possible thing need not exist. Martial arts, science fiction, and animation follow the premises of their imagined worlds. Such premises serve artistic creation perfectly well. Using them as premises about nature, however, gives no assurance that their consequences describe reality. An account must be judged both by the coherence of its reasoning and by the correspondence between its starting point and the actual generation of its object.

## 4 Science and empirical correction

Physics studies nature as it exists. A physical account must therefore do more than make sense: its starting point, reasoning, and consequences must correspond to the world. Physicists select phenomena, specify assumptions and conditions, and express the resulting model mathematically. Mathematics makes the relations explicit and the reasoning open to examination. Observation and experiment determine whether those relations describe the system being studied.

A disagreement between theory and experiment requires the conditions, assumptions, and model to be examined again. Revision is followed by further inference and further tests. Through this repeated comparison, the starting point and its consequences are brought into agreement with the phenomena. The other natural sciences proceed similarly. Chemical reactions and biological observations test accounts of matter and life; strata and fossils constrain reconstructions of Earth's past; astronomical observations test processes beyond the laboratory. In each case, logical reasoning operates within a chosen description, while evidence tests that choice. An inaccurate starting point can sustain a rigorous but increasingly misplaced account.

## 5 Formalisation, mathematics and philosophy

### 5.1 Logic and formalisation

The word "logic" has so far been used broadly. Its specialised meanings distinguish deductive, inductive, and abductive inference, and formal systems such as propositional, predicate, modal, intuitionistic, and many-valued logics. Each addresses particular questions through particular rules. These distinctions are necessary for specialised study, although attention to them alone can obscure the common question of how a conclusion follows from a starting point.

Here logic includes the relations through which things change and the reasoning through which those relations are understood. A starting point specifies conditions and relations; subsequent events or inferences proceed within them. Philosophy, mathematics, and the natural sciences differ in their objects and methods, but all require an account of the connection between premises, processes, and consequences.

Abstraction makes these connections explicit by setting aside concrete details and retaining selected objects, conditions, and relations. Formalisation expresses the selection through concepts, symbols, and rules. Mathematics is the principal language for this work

in the natural sciences. It allows relations to be examined repeatedly and used across otherwise different problems. Abstraction supplies generality; formalisation makes the steps of reasoning precise.

In the *Tractatus Logico-Philosophicus*, Wittgenstein wrote, "The propositions of logic are tautologies" [5, Proposition 6.1]. Such propositions specify formal relations without deciding which physical situation obtains. They supply the structure of inference: given the premises and the rules, the conclusion follows. Their validity does not establish the existence of the objects described by those premises.

Mathematical modelling extends this procedure. Natural objects become quantities and relations; their changes are represented by equations. Once the rules are explicit, repeated steps can be carried out mechanically, including by a computer. This is a major source of mathematical rigour and generality. The same form can describe phenomena whose concrete appearances differ substantially.

The gain has a cost. Constructing a model also selects its logical starting point. Mathematics can establish the consequences of that selection, but those consequences cannot establish its correspondence with reality. The natural sciences can revise a model through observation and experiment. A formal system considered on its own is assessed through the axioms, definitions, and rules with which it began.

Gödel's incompleteness theorems expose a further limit [6]. An effectively axiomatised, consistent system sufficiently strong to express elementary arithmetic cannot capture all arithmetical truth in its theorems. Truth and provability within a specified system are different notions. Formal reasoning remains indispensable, but no such system exhausts truth merely by applying its own rules.

### 5.2 Mathematics and its starting points

Early mathematics grew largely out of practical relations. Counting led to natural numbers; distribution and exchange to arithmetic; surveying, architecture, and astronomy to geometry. Algebra treated unknown quantities, while trigonometry served measurement, navigation, and the determination of celestial positions. Motion, planetary orbits, heat, fluids, and waves later drove the development of calculus and differential equations. In Newton's equations, the heat equation, and the wave equation, an observed process precedes its mathematical abstraction. The starting point retains a recognisable connection with nature.

Axioms are essential to this development. They state the objects and operations on which reasoning depends. Different axioms can describe different natural structures: Euclidean geometry treats flat spaces, while non-Euclidean geometries treat curved ones. Complex numbers, groups, Hilbert spaces, and topology have likewise become tools of electromagnetism, quantum mechanics, particle physics, and the theory of phases of matter. The degree of abstraction alone does not determine whether a mathematical structure describes reality. Its relation to natural structures does.

The important question is how broadly the starting point admits objects, operations, and limits. A narrow description covers part of reality and can be extended when further

relations are needed. An overly broad one also admits objects that nature cannot generate. Once admitted, they possess the same formal standing as objects with a natural counterpart. Correct reasoning then preserves the original overextension: the system treats a merely permitted possibility as an admissible existence.

Fourier analysis illustrates the issue. It arose from the study of heat conduction, representing complicated variations through sines and cosines, and later entered acoustics, electrical theory, communications, and imaging. Measured signals are available over finite times and at finite precision, with finite accessible energy. Engineers usually retain the frequencies needed for a specified accuracy and work with the convergence properties of that particular signal. Mathematics, when extended to arbitrary functions, must also consider convergence in different senses, pointwise oscillations, and objects without a measurement or generation procedure. A method rooted in nature can thus be required to answer questions about a much broader world of formal possibilities.

A differential equation similarly represents a physical process together with its initial conditions, boundary conditions, energy constraints, and specified object. Extending those conditions to arbitrary data or arbitrary notions of solution admits many additional mathematical behaviours. The equation and the deduction can remain correct throughout. What has changed is the range of starting points admitted into the problem.

The Navier–Stokes equations provide a familiar example. Engineers calculate flows in pipes, around aircraft, and in the oceans without first waiting for a theorem covering every admissible initial state for all time. Their problem concerns a particular process, scale, accuracy, and time interval. The mathematical existence and smoothness problem asks whether smooth three-dimensional solutions persist for the full prescribed data class, or whether a finite-time singularity can occur [7]. It therefore covers every possibility admitted by that starting point, including, from the generative standpoint developed here, possibilities that a real preparation may never produce.

### 5.3 Generative limits and mathematical freedom

Mathematicians seek freedom to explore the consequences of clearly stated premises. Given consistent definitions and valid deductions, an object or theorem is accepted within the resulting system. ZFC set theory provides a particularly broad foundation for such exploration. Internal validity establishes a relation to the axioms; it does not establish that every object they permit can arise in nature. Non-measurable sets obtained using choice, Banach–Tarski decompositions, uncomputable functions, and objects given only by nonconstructive existence arguments illustrate the distinction. The criticism concerns the breadth of the admitted starting point, rather than an error in deduction.

This conception of existence has also been contested within mathematics. Brouwer's intuitionism and the constructive traditions that followed give construction a central role [8]. A method for producing an object matters to the meaning of its existence, and some classical forms of inference are consequently rejected. The generative question goes further: from what starting point does the construction proceed, and do its objects and operations correspond to natural change? A construction requires an account of its origin as well as a sequence of valid steps.

The distinction proposed here cuts across the divisions between ancient and modern mathematics, pure and applied mathematics, or one axiom system and another. It concerns the starting point and the conditions under which its objects can be generated. Highly abstract mathematics can describe nature when its relations have a natural counterpart. An excessively broad starting point also supports structures with no such counterpart. Those structures belong to the creative exploration of mathematical possibilities, or mathematics as art.

Mathematics as art can be an intellectual toy: a structure to examine, manipulate, and appreciate. That value does not confer a role in explaining nature. Without a natural correspondence, elegance and internal consistency cannot establish what the world is like. Confusing the two can divert inquiry into increasingly elaborate formal constructions whose refinement conceals their distance from the original physical question.

String theory provides an example of this concern. Many string models introduce ten-dimensional spacetime and supersymmetry, with the additional dimensions compactified on spaces such as Calabi–Yau manifolds. Different geometries, topologies, and parameters admit many possible models. The mathematics may be sophisticated and internally consistent, yet the generative origin of these choices remains the question. Increasing the range of possibilities does not by itself explain why nature realises the particles and interactions that are observed.

Calabi–Yau manifolds have mathematical value, and lower-dimensional instances also enter descriptions of physical systems. The question concerns their use as a starting point for the world: why these extra dimensions, why this compactification, and why the resulting matter content? An elaborate mathematical structure can establish what follows from these choices without explaining why nature makes them. To explain the actual world, the starting point and the structures it permits must themselves be related to natural generation.

### 5.4 Philosophy and the choice of a starting point

Philosophy faces the same difficulty. A system may begin from being, Idea, matter, spirit, the self, or transcendental categories, and develop its consequences with considerable consistency. Its ability to organise its own concepts does not establish its correspondence with the world. Without the direct experimental correction characteristic of the natural sciences, an overly broad concept can be extended until it appears to explain everything while distinguishing little. Such a system may have value as an art of speculation, but that alone does not make it an account of reality. Philosophy must also ask how its concepts arise, which levels they describe, and where their explanatory use ends. Its strength depends on a real starting point for its reasoning.

Giving logic priority need not mean accepting logic alone as sufficient. In Ogden's translation, Wittgenstein wrote, "Logic precedes every experience—that something is so" [5, Proposition 5.552]. The conditions of reasoning do not decide what actually exists. For the purpose of knowing the world, mathematics must account for how its objects arise, and philosophy for how change and relation form existence. This calls for mathematical methods that retain generative origins and philosophical inquiry that follows generation.

Other formal constructions and conceptual systems retain their freedom as mathematics and thought pursued as art.

Social organisations offer a practical example. An organisation is a network of relations among people. Morality, laws, and regulations constrain conduct and sustain order, but cannot substitute for a suitable starting point. With a mistaken starting point, even a refined code can organise action in the wrong direction. Unrestricted democracy and freedom, detached from a common purpose, can impede the formation of a common will; concentration of authority alone can suppress feedback and initiative. An optimal organisation may therefore need two connected starting commitments: "serving the people", understood as serving all its members' common and long-term interests, and the unity of democracy and centralism, through which a common will is formed and put into practice. Democratic participation brings genuine views into decisions; coordinated action gives those decisions effect. Together they sustain individual vitality and collective order.

## 6 Abstraction, levels and emergent reasoning

A natural object contains several levels of order. Atoms form molecules; molecules participate in cells; cells form tissues and organs within a living whole. Each level has characteristic units, spatial and temporal scales, tendencies of change, and laws. The regularities studied by the natural sciences usually concern one or a few such levels.

Abstraction selects those levels for analysis. Fluid mechanics uses density, pressure, temperature, and velocity in place of individual molecular trajectories. Biological models may select genes, cells, organs, or neural networks. Explicit variables and relations make mathematical treatment possible. Reductive analysis gains much of its power by isolating a definite problem from the rest of an object's structure.

The same selection removes information. Basic units at the chosen level are normally taken as given, with their own formation left outside the model. A fluid is treated as a continuum, averaging over molecular motion. A molecular model may take atoms as units while hiding electronic, nuclear, and deeper structure. A neural-network model may reduce neurons to signalling nodes, compressing cellular organisation, molecular reactions, and metabolism into a few parameters. Such models can be accurate at their selected level while omitting the relations that generate and sustain it. Emergence becomes difficult precisely when the higher-level structure is retained but its lower-level formation has been discarded.

Modelling therefore establishes a logical starting point. The basic units, observation scale, variables, and admitted relations are selected together. A parameter change can correct some discrepancies. Others reveal that the objects or scales themselves were poorly chosen. In that case, further calculation within the original model is insufficient: the abstraction and its starting point have to be reconsidered.

Cross-level problems require such reconsideration. When a new level of order appears, the variables used for its constituents may no longer express the laws of the whole directly. The relevant band in the spectrum of laws has changed. Renormalisation and coarse-graining address changes of scale by absorbing short-range degrees of freedom into effective

variables [9]. The derivation of macroscopic equations through kinetic theory by Deng, Hani, and Ma likewise links particle dynamics to statistical descriptions at appropriate scales [10]. In each case, the new description has a starting point related to, but different from, that of the underlying dynamics. Emergent and dialectical reasoning concerns this relation between levels; formal reasoning then develops the consequences within the selected level. A change of scientific paradigm can involve precisely such a reorganisation of the starting point.

A model may select one or a few levels, whereas the natural object exists through them together. Lower-level relations sustain higher-level structures, and higher-level constraints act on lower-level changes. A tree is simultaneously a molecular and cellular process and a whole composed of roots, trunk, branches, and leaves. Its present structure also bears the history of seed, soil, sunlight, seasons, and growth. A rock has atomic and mineral organisation together with traces of pressure, temperature, sedimentation, and crustal movement. These are inseparable aspects of the same existence.

The object thus carries generative relations that exceed those retained in its model. Abstraction is indispensable, but its omissions must not be mistaken for absences in nature. Neither a natural science nor a philosophy can recover the whole world merely by extending deductions from a particular local starting point. Formal reasoning does not automatically restore the levels and relations removed when that point was selected. A theory of everything would have to account for the common origin from which the different levels, structures, and relations are generated.

## 7 From local starting points to a common origin

This concern motivates the search for unification in physics. The Standard Model describes elementary particles and the electromagnetic, weak, and strong interactions; general relativity describes gravity and dynamical spacetime; cosmology concerns the universe's evolution. Logical development and empirical tests establish their use within specified ranges. Each nevertheless begins with assumptions: gauge structure, fields, and parameters in the Standard Model; physical principles and spacetime structure in relativity; an initial state in cosmology. These are local starting points for different parts of the spectrum of laws.

The relation between symmetry and complexity suggests how they might be connected. Towards the foundations of nature, symmetry increases and the laws become simpler. Successive symmetry breaking distinguishes possibilities and permits more elaborate structures. If nature is a continuously generated whole, the starting points of its theories should be traceable to a common, simpler origin. Unification would then explain how the assumptions, objects, and laws of the different descriptions arise together.

The same question extends to mathematics and philosophy. Mathematical structures that describe nature have objects, axioms, and operations whose natural origin also calls for explanation. Philosophy must account for the source of the concepts with which it describes the world. These disciplines use different languages to address the same reality. A common generative account would connect the starting points that make their descriptions possible. We call this the natural starting point.

Finding that starting point would give a definite meaning to a theory of everything. Logical relations would follow the formation of things: their generation would explain why they have the structures they do, why those structures persist, and why their changes are intelligible. Accord with logic, natural generation, and existence as a multi-level ordered structure would then be connected within the same account.

## 8 Disorder and change as the generative starting point

What could serve as this common starting point? A static substance leaves its own change unexplained. Whether the proposed substance is matter, spirit, Idea, or a mathematical object, one must still ask where its rules of change originate. Rules supplied externally leave a prior starting point; placing every rule inside the substance merely restates the world in another form. Formal axioms face the corresponding question of their own origin. The limits identified by incompleteness also prevent us from equating all truth with the consequences of one sufficiently strong formal system. A common generative starting point must account for the formation of objects and rules themselves.

We conjecture that this starting point is disorder, or change itself. Disorder here precedes the relations among already existing things. Change is the process through which things come to exist. At the lowest level, energy is the capacity for change. Before an established structure selects a relation or direction, change is unbiased and open to possibility. It has maximal symmetry and corresponds to what the book calls non-existence.

Relations constrain this possibility. As they repeat, select, and close, ordered structures form and persist. Successive symmetry breaking differentiates changes, establishes units, and permits higher-level structures. Matter, life, and consciousness belong to this continuing process. In the book's formulation, "Disorder creates possibility; order determines direction." The axioms, assumptions, and models of particular sciences are local starting points within that order; disorder and change provide the proposed common origin.

This conjecture gives several research problems a shared starting point. Physics can ask how interactions, gravity, and a Yang–Mills mass gap arise from generated relations. Mathematics can study the formation of natural numbers, complex numbers, and functions, and seek a generative approach to the Riemann hypothesis. Retaining the history of an object's generation also changes how the relation between construction and verification is posed in the $P$-versus-$NP$ problem. Philosophy can examine non-existence as change before persistent relations, and existence as their closure into multi-level order. The three disciplines would then approach their questions through different aspects of the same process.

## 9 Seeking truth from facts

The book develops this argument across elementary particles, spacetime, entropy, life, consciousness, social organisation, and artificial intelligence. These examples serve a methodological purpose: they expose a proposed common logic to comparison across disciplines

and levels. The relation between disorder and order must be examined both in microscopic change and in the formation of living, cognitive, and social structures. Agreement across distant cases gives grounds for pursuing the proposed starting point; it also makes discrepancies easier to locate.

The book expresses the distinction between a description and its object through the metaphor of a finger pointing to the moon. Mathematical forms, philosophical concepts, and scientific models are means of directing attention to nature. Their value depends on what they point to. Asking about the logical starting point means asking where the pointing begins and whether its direction is grounded in the object's actual generative relations.

The specialised examples are necessary for this comparison. A plausible argument alone cannot establish that one method applies across levels, nor can microscopic examples establish its use for life or cognition. The recurring questions remain concrete: where does the reasoning begin, over what range does it apply, what has been omitted, and does its conclusion correspond to the object? An argument valid within one domain can fail when carried beyond it.

Seeking truth from facts begins with what is real. It requires revising an account in response to the facts rather than selecting facts to support a conclusion already accepted. Each new problem calls for its object, level, conditions, and range to be identified afresh. A relation established at one level cannot simply be transferred to another without examining how the two are connected.

A persuasive claim therefore deserves neither immediate acceptance nor immediate rejection. Its premises, domain, and omissions can first be examined. A coherent route need not lead to the intended destination. To see a path without being bound to it is to preserve the ability to compare the argument with reality and revise its starting point. The following appendices explain how this method guides research into physical structures, mathematical objects, and engineering processes.

## A The physical picture: from generated spectra to unification

The physical programme asks how a changing world acquires structures that persist, interact, and admit laws of their own. A common origin must account for the relations within a structure and for the changes by which it communicates with its surroundings. Internal geometry, particles, fields, and collective motion enter as successive forms of this organisation.

### A.1 Source, carrier and internal geometry

A particle theory usually specifies its fields and gauge group before studying their dynamics. The generative programme follows the formation of those starting data. Minimal unbiased compositional change first supplies a source and a carrier on which changes can be compared. The least-dimensional nontrivial compact connected faithful irreducible orthogonal carrier is a rotation plane. Independent composition, exact same-shape scaling, and unit

covariance determine square-root scaling and the centred Gaussian law within the stated composition class [11, 12].

The Gaussian describes how correlation changes with separation. Its local loss and total correlated extent give two lengths calculated from the same kernel. Constant preservation selects the reflecting closure within the separated homogeneous local boundary class. The resulting finite window supports distinguishable spectral modes. Thus a spectrum enters through a definite account of how changes remain related and how their response is observed.

The physical continuation also compares successive operations. Ladder operations and coherent comparisons generate the round $SU(2)$ process spectrum. Complete observations invariant under the centre give the quotient $\mathbb{RP}^3$ [12]. The rotation plane describes the minimal change carrier, while this internal geometry organises relations among its processes. Each step concerns an explicit action and the observations that preserve it.

This gives internal geometry a physical role. It specifies which changes can be distinguished, which comparisons are equivalent, and how a retained mode responds to a source. The space, its spectrum, and its observation algebra are therefore constructed together. Particle labels acquire meaning through their place in these actions and responses.

### A.2 Gauge closure and the Yang–Mills mass gap

Different descriptions of the same internal relations must give the same observations. Gauge transformations express the permitted changes of frame. Adjoint, exchange, source phase, and reversal constrain how the generated relations continue into gauge and matter sectors. In the faithful continuation studied in [12], these requirements select the gauge group $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, the matter module, and a complex scalar doublet linking the sectors. The scalar has a structural purpose, since relations between sectors require a channel through which their responses can be compared.

Mass then concerns the cost of changing a persistent structure. A configuration can have internal activity while its centre remains at rest. Its excitation energy must be read from the same response and normalization that define the configuration. For the colour connection, the source supplies an equilibrium problem. Radial variation changes its collective amplitude. Common-frame coexact variation changes the retained connection directions. The equilibrium curvature supplies a radial restoring cost, while the internal spectrum supplies the cost of the coexact modes.

Their source-normalised combination defines the mass operator. The generated Yang–Mills mass-gap theorem establishes a strictly positive first excitation above the vacuum in the complete equilibrium response [12]. Its value is the smaller of the radial and coexact thresholds. The vacuum is stable, and a centred excitation has a positive cost fixed by the generated operator. The internal scale and its physical calibration determine the units in which that cost is expressed.

The vacuum representation carries the original connection state, its polynomial observations, and its update. Consequently the spectral gap controls observations of the same connection whose equilibrium was constructed. Reconstruction on the event carrier then

connects these fluctuations with local fields. The research programme follows this relation between source, vacuum response, and field description through the relevant resolutions.

### A.3 Stress response and collective gravity

Gravity asks how the organisation of matter can produce a common long-range response. Energy and momentum provide the natural source. The transverse traceless tensor channel isolates a spin-2 response from the scalar and longitudinal parts. A pole at zero invariant mass, with nonzero coupling to the matter source, gives a collective mode that can be excited at arbitrarily long distances [13].

The coupling is essential to this picture. The response includes the source map, propagation, observable readout, and any direct term. The pole residue measures how strongly matter excites and observes the mode. Coarse-graining transports these data together, with controlled changes in the visible pole and residue [14]. This connects the spectral description to an interaction between actual sources.

The full local stress tensor and the reduced internal tensor response obey different spectral constraints. The reduced response is the channel in which the collective pole is studied. For a two-helicity massless spin-2 response with universal matter coupling, soft-limit constraints and local consistency give the Einstein–matter form at leading long distances [15, 16]. The route to effective geometry therefore passes through a source-visible mode and its coupling. The programme asks how that response is sustained and transported as the organisation of matter changes.

### A.4 Couplings, scales and cosmic evolution

A common internal spectrum also connects particle content, interaction strengths, and mass scales [17]. Source normalization sets how responses are compared. Spectral capacity counts the participating directions, screening changes their effective response, and scalar closure links the sectors. Matching then relates the description at different resolutions.

The resulting constraints connect quantities often treated independently. Source and colour screening select three matter copies. A shared response family has a selected shell level and a unique positive endpoint root. Eliminating common transfer parameters gives relations among geometric fermion masses, while the same spectral normalizations enter electroweak and cosmological comparisons [17]. These joint dependencies give the construction its empirical content. Changing a shared response changes several outputs together, so each comparison tests more than an isolated parameter choice.

At larger scales the useful units become matter concentrations and their collective motion [18]. A distant interaction can be described through mass and spatial moments. During an encounter, internal field information becomes relevant again. Passing between these levels requires the observable force to be preserved and the memory of unresolved directions to be carried forward. Cosmic structure formation is then a study of how new effective objects emerge while remaining connected to their field description.

Unification is pursued through these connected levels. Gauge relations organise internal comparisons, mass records the cost of excitation, gravity concerns a common collective response, and cosmic evolution changes which structures provide the useful description.

Their proposed common origin becomes a scientific question through the transformations that relate them and the observations those transformations preserve.

## B The mathematical picture: generated objects, scale energy and complexity

Retaining generation changes what can be asked about a mathematical object. Its value is accompanied by operations, relations, and a history of formation. These data make it possible to compare constructions that agree at their endpoint and to study how their differences affect later operations.

### B.1 Arithmetic and the generation of its objects

The arithmetic construction begins with the same compositional change and Gaussian correlation source as the physical construction. Local and total correlation lengths determine the finite window, and constant-preserving reflecting closure gives an ordered spectrum. The modes are first ordered by their response. On their oriented analytic lifts, pointwise modulation and map composition define native addition and multiplication. Modulation by the first positive mode defines successor. Winding degree subsequently certifies the natural-number arithmetic carried by these operations [11].

The construction also retains how an object was formed. Pure addition of six units and multiplication of a sum of two units by a sum of three units have the same value. Their ordered trees have six and five leaves, respectively. Equal root value and equal Gaussian root energy therefore coexist with different internal histories. The expression resolution keeps these histories separate until algebraic evaluation identifies their common value.

The source assigns equal conditional weights to the available root channels and assembles their child histories independently. The resulting positive history law, weighted by the Gaussian root mass, recovers the original value mass under marginalization. A source-weighted Hilbert readout separates the value component from history differences invisible at that readout. The complete assembly response couples these components, and an explicit nonzero matrix element proves that a retained difference can affect a later value observation [11]. Eliminating the difference component gives a value equation with memory. The distinction between a number and its formation thus enters an actual evolution on the associated history space.

Prime decomposition provides another view of the generated relations. Multiplicatively irreducible modes determine the prime directions. Their residue classes form a basis of the augmentation cotangent space, and the corresponding one-word vectors form the degree-one Hodge kernel. The decomposition heat flow isolates this prime subspace quantitatively. Symmetric and exterior occupation spaces then organise factorization, square-free selection, and signs on spaces built from the same natural-mode family [19]. Arithmetic can consequently be studied through its operations, its formation histories, and its operator responses.

### B.2 Functions, operations and limits

For functions, the corresponding question is what survives the operations that form a new function. A generated class specifies source objects, permitted operations, and controlled limits. Linear combination, composition, transport, integration, and projection carry both an object and the estimates required for its subsequent use [20].

Nonlinear evolution shows why this matters. Interacting modes create new frequencies, so an initial decay estimate must be followed through the interaction. Gaussian decay at the source and analytic decay preserved by a nonlinear operation play different roles. The useful spectral weight is one whose estimate propagates through the operations actually used in the construction.

An infinite object is then approached through finite presentations. A stronger spectral bound controls the unresolved tail, while a weaker observation norm determines convergence. This separates the information needed to form the limit from the accuracy required to observe it. The resulting function class records which constructions and limits retain those estimates. Existence, regularity, approximation, and observation can be examined along the same sequence of operations.

### B.3 Scale energy and the Riemann hypothesis

The Riemann hypothesis brings a particular difficulty into focus. Multiplicative generation determines the Möbius signs, while an additive spectral observation combines many labels at a common scale. The divisor inverse relation fixes the coefficients. Gaussian scale transport turns them into a signed response whose quadratic energy contains the correlations between different labels [21]. Those correlations are part of the quantity to be understood.

The Mellin transform connects scale evolution to the reciprocal zeta function. For the signed Gaussian shell, explicit transform factors relate a scale estimate to the domain of holomorphy and hence to zero-free regions of zeta. In the weighted-energy formulation, uniform bounds on the prescribed finite quadratic forms for every fixed $1/2 < \sigma \leq 1$ are equivalent to the Riemann hypothesis. The bound may depend on the weight, while remaining uniform as the observation scale increases. Explicit tail bounds connect these finite observations to the complete response.

The research programme seeks that uniform estimate from the common divisor source. Prime occupations, common factors, expression relations, and additive organisation provide different descriptions of the same coefficients. Finite Fourier and Weil coordinates can expose their relations within a scale observation. Their usefulness is measured by what they retain of the complete signed response. Taking absolute values separately loses precisely the cross terms through which cancellation can occur. The central task is to use the generating relations together before making the estimate.

### B.4 Generation, verification and $P = NP$

A description of an answer may conceal the process that produced it. This distinction becomes important when finding an answer appears much harder than checking one. A generated computational task specifies its source data, permitted operations, and requested

observation. The calculation retains a finite record of how it reaches that observation. The cost of forming the record belongs to the computation.

Certified spectral evolution gives a concrete realization of this idea [22]. For the specified semilinear Neumann input family, rational operations construct a finite trajectory with residual and error bounds, and an independent procedure checks the record. Construction has a polynomial bit-cost bound in the complete input and precision parameters. Verification has a polynomial bit-cost bound in the input and record lengths. Intermediate number sizes and the accuracy of the final observation enter these estimates.

The broader formulation considers relations equipped with a uniform certified route from input to a successful witness. Permitted operations, polynomial clocks, decoding margins, and resource bounds are retained in that route. In the resulting framework, languages with certified relation presentations coincide with languages decided by its permitted deterministic polynomial-time procedures. Efficient generation and verification are compared through an explicit construction rather than through the endpoint description alone.

The connection with the classical $P$-versus-$NP$ question concerns which verification problems admit such routes [23]. If every standard $NP$ language has a verifier with a uniform polynomial certified route, the construction gives $P = NP$ in the standard bit model. This identifies the role of the generative programme. It seeks source relations and update laws that produce successful readouts with controlled total cost, and studies how widely those constructions extend across finitely encoded problems.

## C The engineering picture: finite constructions and continuum dynamics

The engineering question is how a finite calculation remains attached to a continuously changing process. A useful model must explain which structures it resolves, how they exchange activity, and what changes when the calculation requires a finer description. Generation gives these questions a common setting.

### C.1 Preparation, evolution and observation

A flow begins with a physical preparation, a geometry, imposed forces, an observation scale, and a time interval. These data determine what the calculation is asked to describe. In the generative approach they remain attached to the evolving state and its observations [24]. Resolution is selected in relation to that process and its requested accuracy.

Viscous incompressible dynamics combines two different tendencies. Stokes transport damps fine variation, while nonlinear advection transfers activity between modes. Leray projection preserves incompressibility. Spectral projection and time integration turn these operations into finite trajectories. Following their composition makes it possible to compare each trajectory with the original equation, including its initial discrepancy, full residual, unresolved tail, and stability.

These estimates explain what a finite state represents. A small retained component can coexist with appreciable activity outside the current window. When that activity affects the required observation, the description must change. Refinement follows the evolving process and the accuracy of its readout.

### C.2 A constructive route to Navier–Stokes existence

For the periodic incompressible problem, finite Fourier preparations give explicit starting states. Viscous smoothing and nonlinear transfer then act on the same generated trajectory. The continuous solution is approached by refining these finite constructions while carrying their estimates through each refinement [24].

The crucial link is between residual and stability. A residual measures how far a trajectory departs from the equation. Stability controls how that departure affects subsequent states. When the full residual tends to zero and a common stability estimate holds on the chosen interval, the trajectories form a Cauchy family. A common weighted Stokes estimate controls the high-frequency tail and supplies the additional integrability needed for uniqueness and positive-time regularity. Completing the family then produces a regular continuum solution.

For finite Fourier data with zero forcing, the construction supplies a common local interval on which these estimates can be established. Continuation concerns the propagation of the same resources to later times. The question of persistent regularity can thus be followed through the competition between viscous smoothing, nonlinear transfer, and spectral concentration. Extending those controls across the prescribed data and time ranges connects this construction with the classical three-dimensional existence and smoothness problem [7].

This gives a concrete meaning to existence through generation. The finite operations, their convergence, and the regularity of the limit belong to one account. A continuum field remains connected to the preparations and estimates through which it was obtained.

### C.3 Adaptive models and reusable calculations

Predicting a state and explaining its transfers require complementary information. Viscous loss, nonlinear transport, forcing, and approximation enter quadratic balances with different signs and cross terms. Two calculations may have similar state accuracy while resolving those transfers differently. State-error and quadratic-balance certificates therefore accompany one another [24]. Their joint control makes the calculation useful for both prediction and diagnosis.

Consider a flow developing concentrated activity from a smooth large-scale preparation. Its spectral window can expand when finer modes affect the observation, then contract as viscosity suppresses them. At a change of window, residual and tail bounds retain the connection to the equation. If unresolved variables influence later motion, their effect enters through retained memory. The same issue arises when a distributed field is replaced by objects described by mass and moments. Such a change of level is governed by the response that must be preserved [14, 18].

Finite records make these changes inspectable and reusable. Construction, residual evaluation, and independent verification can be compared in a common finite encoding [22]. A calculation can then be refined, extended, or confronted with new observations while carrying the information that supported its previous use.

This is the proposed engineering role of the framework in adaptive simulation and digital twins. A model evolves with its physical preparation, its retained scales, and the accuracy required of its observations. New computational methods, including learned proposals, can enter this process through their equation residuals and observable errors. The guiding aim is a calculation whose formation and limits remain understandable as its description changes.

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